

Hybrid Powertrain Systems for 100 kg Multicopters and Tailsitters

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This paper explores the use of hybrid powertrains in vertical takeoff and landing (VTOL) unmanned aircraft configurations using a physics-based conceptual sizing tool. Three multicopter configurations (quadcopter, hexacopter, and octacopter) and their corresponding winged biplane tailsitter variants were sized for 100 kg takeoff mass for mission distances varying from 25 to 250 km. The hybrid powertrain model accommodates power sharing between a battery and an internal combustion engine/generator/rectifier combination. The engine was sized to provide cruise power, while a “boost” battery was sized to provide the necessary additional power required to take off and land vertically. Accommodating two different power sources and their corresponding energy storage systems in a VTOL aircraft affords better cruise efficiency, because the engine is operating at its minimum specific fuel consumption condition, at the cost of additional empty weight. This strategy does not offer noticeable benefits (over powertrains without a boost battery) for multicopter configurations operating in edgewise forward flight because cruise power and hover power are comparable, and peak power for engine sizing is not strongly driven by any one flight condition. For winged VTOL configurations, where cruise power is less than 40% of hover power, the dual-power plant solution enables a twofold increase in payload through both fuel and empty weight savings.

I. Introduction

VERTICAL takeoff and landing (VTOL) aircraft across different weight and size classes have been designed and deployed for applications ranging from border patrol and geographical survey to passenger transport. Some examples of VTOL unmanned air vehicles (UAVs) are shown in Fig. 1. Most of these motor-driven multicopter aircraft are powered by batteries with 150–200 Wh/kg packs; these battery specific energy levels allow roughly 10% of the achievable range of similar aircraft powered by fuel-burning engines. Powertrain hybridization (driving motors using electricity produced in-flight by an engine-generator combination) is key to extending the range of VTOL aircraft with distributed electric propulsion (DEP). Where practical, a low-capacity battery may also be present to provide short-term power boosts for rapid gust rejection and maneuvers or serve as a backup power source for emergency landing following engine failure.

This paper examines the use of hybrid powertrains in a winged tail-sitter package delivery VTOL aircraft using conceptual sizing. Estimating weights and performance of this unique vehicle requires several modifications to legacy rotorcraft sizing workflows, discussed in the following section.

A. Rotorcraft Sizing: State-of-the-Art

NDARC [1] is a comprehensive sizing tool that estimates weights and performance of both fixed-wing and rotary-wing aircraft. Several vehicle configurations can be represented in NDARC, including

(but not limited to) single-main-rotor, coaxial, and tandem rotor helicopters, tilt-rotors, and various fixed-wing aircraft. Using models for component empty weights, lift and drag of lifting and non-lifting bodies, and power plant performance (engine and motor), an aircraft or rotorcraft can be sized and/or analyzed for multiple missions. Dissimilar rotors, wings, power plants, and hybrid powertrains can be modeled and analyzed with NDARC, rendering it the “gold standard” in rotorcraft sizing capabilities.

Sizing workflows typically rely on low-fidelity empty weight models; these models are distilled from empty weight breakdowns of full-scale production rotorcraft. Assumptions of aircraft configuration, load path, locations/type of rotors, and construction material are implicitly included in these scaling laws. While some weight assumptions may be judiciously modified with good confidence (e.g., replacing metals with composites in primary structures), fundamental changes to the aircraft configuration or its load paths may not be well represented using the same key parameters or their scaling exponents. Even with the appropriate aircraft configuration(s), application of the scaling laws to very heavy or light vehicles may constitute extrapolation of trends, and scale-specific considerations (e.g., strength vs stiffness based sizing criteria for wings and blades) are ignored.

Load paths and empty weights are configuration-specific for VTOL platforms featuring distributed vertical lift. Depending on the layout and relative placement of the rotors, different configurations may feature very different scaling laws and driving parameters. Additionally, for vehicles with more than four rotors, the need to land safely (without autorotation) after single rotor/motor failure drives motor and airframe sizing with reduced blade inertia and mass. Blade design for multirotor VTOL platforms represents a marked deviation from the existing helicopter paradigm, because the design drivers are fundamentally different. These unique rotor systems have sizing requirements that closely resemble strength-based sizing and mass buildup rules used for propeller blades. The nature of these design drivers (and lack of available weight databases) necessitates configuration-specific resolution of empty weights and load paths in conceptual sizing—a practice not typically observed or required for conventional helicopter or tilt-rotor sizing. In previous sizing studies for VTOL UAVs using the HYbrid Design and Rotorcraft Analysis (HYDRA) framework, the authors have introduced appropriate higher-fidelity weight and performance

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Fig. 1 Unmanned aerial platforms with distributed vertical lift.

models into sizing workflows—see Govindarajan et al. [2] and Sridharan et al. [3,4]. These improved weight models are adopted for sizing UAVs in the present study.

For powertrain components, weights are sensitive to the sizing torque and power of the motors and inverters, respectively. The presence of different types of power plants requires systematic bookkeeping of power flow for each power source across various mission segments for each sizing mission profile. Both NDARC and PEACE [5] are examples of frameworks that accommodate and analyze vehicles with multiple dissimilar power plants. Recently, Chakraborty [5] surveyed aircraft sizing workflows with battery-electric and hybrid-electric propulsion systems and sized various battery-electric and hybrid-electric passenger-carrying eVTOL aircraft. Hybridization stands to offer similar benefits for unmanned VTOL aircraft; Elroy Air's *Chaparral* is one such example with an advertised range of almost 500 km and a maximum speed of over 120 knots [6].

B. Hybrid-Electric Powertrains for Winged VTOL Aircraft

Package delivery missions for VTOL aircraft present challenging sizing scenarios for the propulsion system (compared to their fixed-wing counterparts). A long-duration cruise dominates most of the operating flight time, bounded by short hover segments for lift-off/touchdown. For hybrid propulsion systems powered by an engine-generator combination, power plants are sized for liftoff hover power (plus margins). When cruise power is noticeably lower than hover power, a portion of the empty weight (associated with the difference between hover power and cruise power) is dedicated to upsizing the engine for vertical liftoff. Engines sized for hover also suffer increased specific fuel consumption (SFC) in level cruise owing to thermodynamic efficiencies peaking at maximum output power; engine fuel consumption in cruise is therefore higher when sizing for hover power. Arace et al. [7] recently showcased the impact of power plant modeling details on vehicle range by integrating thermodynamic and electromechanical efficiency maps from a piston engine and generator/motor, respectively, into rotorcraft sizing.

Sizing the engine for a long-duration cruise segment enables overall vehicle weight reductions. If the engine is sized for cruise power and a battery pack is used to augment power in hover, then empty weight savings may be realized by reducing the size of the engine and generator. This design also features reduced fuel consumption by ensuring a near-constant engine power demand across the mission profile. The key premise in this study is that the battery is a lighter electric power source than a piston engine-generator combination (combined specific power of 0.8 kW/kg); the same metrics do not strongly favor a fast-discharge battery over a turbine engine with a specific power of ≈ 4 kW/kg.

While some studies [8–13] have identified the key benefits of a hybrid-electric powertrain for fixed-wing and VTOL aircraft, conceptual sizing with detailed models of various aircraft components—including powertrain components—is required to identify missions where such hybrid powertrains would provide benefits for VTOL platforms. In this study, we expand the HYDRA framework to

accommodate dissimilar power plants and introduce rigorous management of power share between energy sources in sizing optimization, and we use the expanded framework to study hybridization benefits for a winged tailsitter delivery drone.

C. Objective

The objective of this paper is to study and quantify the benefits of hybridizing the piston/generator power plant with a boost battery by performing conceptual sizing of multirotor and hybrid VTOL configurations with 4, 6, and 8 rotors. Specifically, this study will cover the following:

- 1) Multirotor VTOL aircraft sizing with three power plant combinations: battery-electric, engine-generator, and battery + engine-generator
- 2) Comparison between various edgewise multirotor UAVs and their winged tailsitter counterparts (with different powertrain options) for fixed takeoff mass at various sizing mission distances
- 3) Identification of relative advantages of battery-engine-generator hybrid for various sizing mission distances.

II. Methodology

The conceptual design studies presented in this work were carried out using the sizing tool HYDRA. Details of the framework and models are documented in the authors' previous studies [3,14]. Configurations are "assembled" from a designer-specified set of components (rotors, wings, motors, engines, and batteries), and their respective mechanical and electrical "connections" (mount points for rotors on wings/body, drivetrain matching between motors and rotors, as well as power transmission from power plant to individual rotor). Geometry estimation/rotor placement, aerodynamic performance, weight estimation, and component connectivities are abstracted into the backend, enabling users to size a wide variety of VTOL configurations ranging from single main rotor helicopters to multirotor UAVs and eVTOLs (with appropriate component weight and performance models) with a minimal set of inputs. Vehicle layout and mission assumptions are explicitly specified using both parameters and programmable elements (scaling laws or direct function calls to user-defined models for performance and weight estimation). This flexibility enables the combination of predefined and user-defined models in sizing. These component models may range from zeroth-order methods (parametric fits for existing data) to detailed physics-based models. Sizing is mathematically formulated and implemented as a single-objective Multi-Disciplinary Optimization (MDO) problem subject to bounds and constraints. The following section details the relevant aspects of HYDRA for sizing an aircraft with a hybrid powertrain.

A. SAND: Simultaneous Analysis and Design

Formulating aircraft sizing as an MDO problem enables the systematic separation and resolution of interplay between various subdisciplines (aerodynamics, structures, flight control, and propulsion) while satisfying the consistency conditions in each domain. In the present study, a Simultaneous ANALysis and Design (SAND)

MDO approach is adopted. In SAND, the disciplinary design variables from weight estimation, aerodynamic performance/trim, and high-level sizing are driven by a single optimizer; this formulation is shown schematically in Fig. 2.

In SAND, disciplinary constraints are “promoted” (in MDO parlance) to vehicle-level sizing constraints, allowing the optimizer to drive both the overall aircraft geometry/weight as well as each of the subdisciplines (e.g., airframe tube sizes and rotor trim thrusts in each flight condition) to a system-level optimum. The advantage of SAND lies in its ability to avoid regions where disciplinary constraints may fail (e.g., certain combinations of design variables may result in a vehicle design where the standalone trim process fails to identify a solution). When compared to other approaches (such as full factorial search, Multi-Disciplinary Feasible [MDF], and bilevel MDO), this process converges rapidly but incurs an overhead in initial implementation for problem setup. Because trim is imposed as a set of equality constraints, a gradient-based optimizer was used to drive sizing. In this work, the Sequential Least Squares Programming (SLSQP) [15] optimizer from *Scipy* [16] was used to drive sizing with a SAND formulation. More details are available in the authors’ previous work [17].

To obtain an initial guess for the SAND optimizer, seed inputs for the design variables (e.g., rotor radius, solidity, tip-speed, wing aspect ratio) are provided, and a self-consistent solution is obtained. In the present work, all the designs feature a fixed maximum takeoff weight (MTOW). Subsequently, the design is evaluated using various performance and weight models. In this process of obtaining an initial design for the SAND optimizer, vehicle-level constraints are not enforced. For example, no limit is placed on the rotor blade loading coefficient in any mission segment, so this initial guess with consistent weights may still result in a stalled rotor. Alternately, the initial design may feature excessively heavy components, resulting in a negative payload; such designs are not eliminated at this stage. While these designs are not practical, a “rough” initial guess is sufficient to initiate a follow-on optimization, at which stage all the constraints are enforced to ensure that the final design conforms to various physical limits.

B. Design Variables and Constraints

A list of design variables, their bounds, and constraints for sizing are discussed in this section and summarized in Tables 1 and 2.

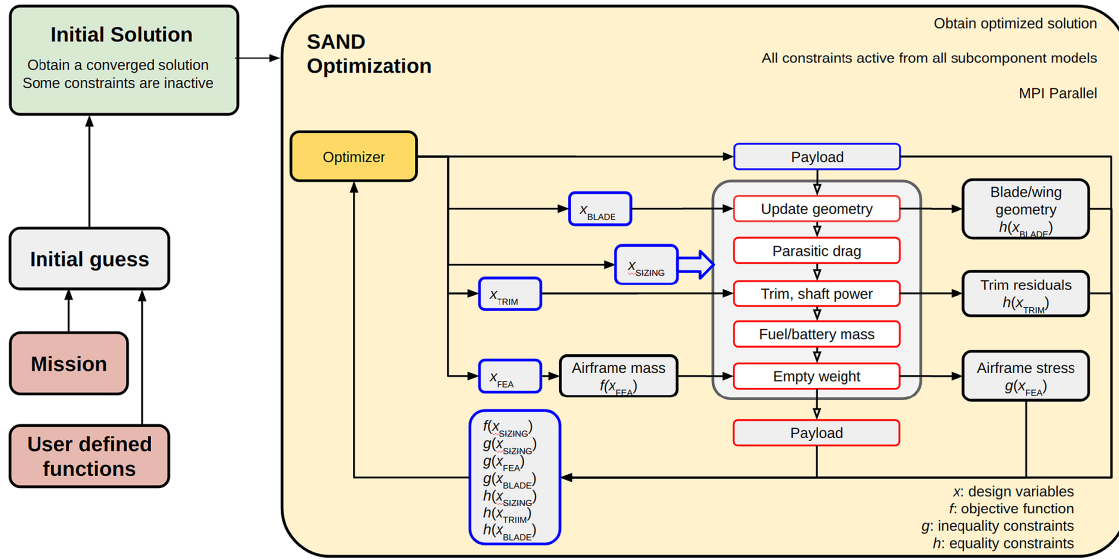


Fig. 2 Conceptual design in HYDRA using a Simultaneous Analysis and Design (SAND) framework.

Table 1 Design variables and bounds for optimization-based sizing in HYDRA

Type	Design variable	Min	Max
Seven sizing variables	Rotor radius, R	0.2 m	1.0 m
	Rotor solidity, σ	0.06	0.2
	Hover tip speed, $V_{TIP} = \Omega_h R$	70 m/s	180 m/s
	Cruise RPM ratio, Ω_c / Ω_h	0.5	1.2
	Wing aspect ratio, AR	3	15
	Power fraction across power plants	0.0	1.0
Objective/“dummy” design variable		Payload mass	
Five FEA variables	Tube size: wing frame d_{WING}	0.01 m	0.5 m
	Tube size: cross-brace d_{XBRACE}	0.01 m	0.5 m
	Tube size: payload bay section frame d_{BAYCS}	0.01 m	0.5 m
	Tube size: rotor mounts d_{MOUNT}	0.01 m	0.5 m
	Tube size: payload bay vertical frame d_{VERT}	0.01 m	0.5 m
$(N_{rotor} + 1)N_{seg}$ trim variables	Rotor 1 thrust/takeoff weight $= T_1 / W_{TO}$	0.0	1.0
	...		
	Rotor N_{rotor} thrust/takeoff weight $= T_{N_{rotor}} / W_{TO}$	0.0	1.0
N_{seg} hybrid power fraction variables	Body pitch attitude	-10 deg	180 deg
	Battery fraction or Engine fraction	0.0	1.0

N_{seg} : number of flight segments; N_{rotor} : number of rotors; W_{TO} : takeoff weight in newtons.

Table 2 Constraints for optimization-based sizing

Constraint type	Description	Number
Payload weight matching	Calculated payload weight ≥ 0	1
Aerodynamics	Blade loading coefficient ≤ 0.12 (all rotors, all flight conditions)	$N_R N_{\text{seg}}$
Airframe	Von Mises factor of safety ≥ 1.4	$N_B N_{\text{seg}}$
Trim	Translational acceleration = 0	$2N_{\text{seg}}$
	Rotational acceleration = 0	$1N_{\text{seg}}$

N_{seg} : number of flight segments; N_R : number of rotors; N_B : number of unique beams.

These design variables may be grouped into various categories based on their respective roles in sizing; the treatment of these different types of design variables is elaborated below.

1. High-Level Sizing Parameters

The vehicle sizing parameters considered in this study are listed in Table 1, together with their upper and lower bounds. The high-level sizing parameters fall into three categories: rotor, wing, and power plant.

Rotor: The sizing variables for a rotor are its radius, geometric solidity, hover tip speed, and cruise tip speed (specified through an RPM ratio, i.e., the ratio of rotor RPM in cruise to rotor RPM in hover). All rotors are assumed to be two bladed and provide vertical lift in hover and propulsive thrust in cruise. The choice of two-bladed rotors is driven by two requirements:

- 1) achieving a sufficiently high Reynolds number on the rotor airfoil sections, and
- 2) obtaining sufficient rotor blade cross-section thickness to improve vertical bending stiffness and torsion stiffness.

Wing: The winged VTOL platforms in this work feature tractor rotors placed ahead of the wing leading edge. Wing span scales with rotor radius as $1.5N_R$, which is the spanwise separation of two rotor hubs, illustrated in Fig. 3. Wing chord is obtained from wing span using aspect ratio as a free design variable.

Power plant: For hybrid powertrains, the fraction of power supplied by the battery in different mission segments are design variables. The power fraction (χ) is defined as

$$\chi = \frac{\text{Power supplied by battery}}{\text{Power supplied by engine} + \text{Power supplied by battery}} \quad (1)$$

Peak engine power demand drives engine/generator sizing. Battery packs are sized using a pack specific energy of 158 Wh/kg. This relatively low value accounts for degradation and unusable margins associated with low voltage cutoff criteria at high depths of discharge.

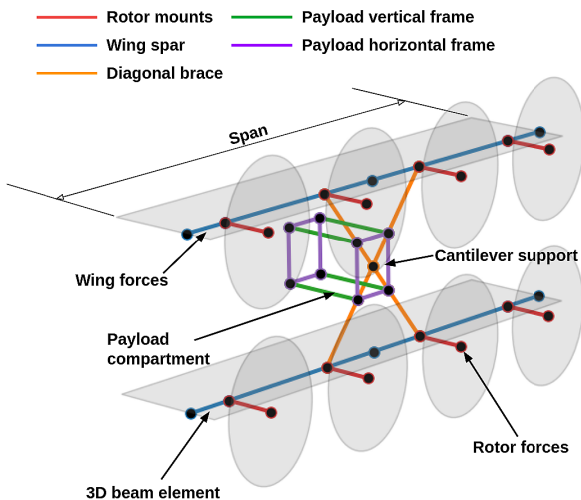


Fig. 3 FEA model of example 8-rotor tailsitter airframe in HYDRA.

A high-level inequality constraint is imposed to guide the sizing to realistic configurations: A maximum limit is imposed on the rotor hover blade loading coefficient to ensure adequate stall margins ($C_T/\sigma \leq 0.12$).

2. Coupling Variables (System-Level)

To aid convergence, coupling variables (values passed between disciplines) are also promoted to the vehicle level sizing optimizer in SAND, and equality constraints are added to resolve the cyclic nature of weight dependencies without resorting to successive substitutions. In the present study, payload mass is added as a coupling variable to be driven by the vehicle-level optimizer; more details are provided in Ref. [17].

3. Coupling Variables (Disciplinary):

Similar to the treatment of payload mass, vehicle trim variables (in each mission segment) are also promoted to the system-level sizing optimizer, and trim equations are treated as equality constraints for vehicle sizing. The trim variables for a given flight condition are the N_R rotor thrusts and body pitch attitude. These trim variables and their bounds are shown in Table 1. The three longitudinal trim equation residuals are normalized to obtain the residual linear and angular accelerations; the vehicle is trimmed by driving these acceleration residuals (in each flight condition) to zero using $3 \times N_{\text{SEGMENTS}}$ equality constraints.

4. Airframe Design Variables

The airframe (including wing spars) is modeled as a lattice of beam elements with hollow circular cross-sections. A finite number of tube sizes are allowed to reduce the count of unique parts. The airframe design variables are the outer diameters of each type of hollow tube. For airframe sizing, inequality constraints are used to ensure that material factors of safety based on Von Mises stress are within an allowable limit $\sigma_{\text{VONMISES}} \leq \sigma_{\text{MAX}}$ for each of the beam elements in sizing flight conditions (hover and outbound cruise with payload), yielding 30×2 inequality constraints. Aluminum construction is assumed for the tubes.

C. Component Models in HYDRA

1. Empty Weight and Battery Weight Models

Vehicle empty weight was accumulated from the buildup of weights for the following components: 1) rotor blades, 2) fixed wings (skin and foam) and airframe structure (including wing spars), 3) brushless DC electric motors and controllers, 4) generator and AC/DC converter, 5) battery, 6) engine, and 7) miscellaneous parts (payload bay shell structure, avionics, onboard computers, fairings, fasteners, nuts/bolts, onboard sensors, hubs, and glue). Figure 4 shows the schematic of the hybrid engine-battery powertrain with all the mentioned components.

The treatment of entries in the list of remaining empty weight components is discussed in this subsection.

1) **Rotor blade weight model:** The rotor blade comprises both load-bearing components (spar, skin, and root fitting) and non-structural masses (paint, glue, and balance weights). Masses of load-bearing components were calculated based on the vertical bending and shear loads due to lift, as well as centrifugal forces due to blade rotation. Effects of precone on root bending moment

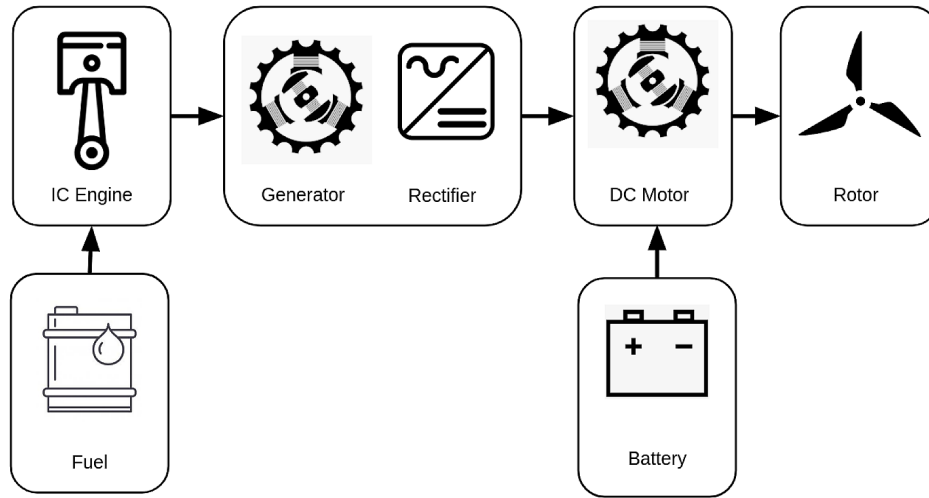


Fig. 4 Schematic of the hybrid engine-battery powertrains.

relief were included. The key advantage of this model lies in its formulation and solution procedure; no additional design variables or constraints are exposed to the rest of the sizing process because spar dimensions (and mass) are determined based on a semi-analytic solution of nonlinear equations in one variable (i.e., equality constraints were recast as equations and solved directly). The complete derivation of this model is included in the authors' previous work [3].

2) **Wing and airframe weight models:** Wing skin weight was estimated assuming minimum gauge construction (two layers of bidirectional carbon fiber) with a foam filler material. Owing to the smaller geometric scales for this vehicle (0.5–1.0 m radius), structural dynamics/resonance concerns are ignored. Cross-sectional shear and bending moments are carried by the spar. For the configurations investigated, wing spar sizing and mass estimation was integrated into airframe weight estimation through a finite element analysis (FEA)—the lattice structure of which is shown in Fig. 3 for a representative octarotor biplane. The methodology largely follows the formulation outlined by Sridharan et al. [3].

3) **Motor and speed controller:** Motor and ESC weights are estimated, respectively, using specific torque (10 Nm/kg) and specific power (28 kW/kg) based on commercial off-the-shelf components from T-motor (specifically, the U11 II, U13 II, and U15 II “Power” motors, and Thunder, Alpha 120, and Flame 60A ESCs [18]). A combined motor + ESC electromechanical conversion efficiency of 86% was assumed. ESCs were sized to accommodate 150% of maximum trim power to allow for transient currents necessitated by gust rejection and maneuver accelerations. Motors were sized to provide 131% of hover trim torque, corresponding to 150% maximum hover power.

4) **Generator and rectifier:** Generators were sized based on a specific power of 4 kW/kg. Rectifiers were sized assuming a specific power of 2.5 kW/kg, based on the aviation-grade

IPS3000 and IPS6000 series commercial off-the-shelf components advertised by Intelligent Power Systems [19].

5) **Battery:** The mass of the battery was calculated using the mission pack energy output and a pack specific energy of 158 Wh/kg. Discharge efficiency losses due to high C-rates, capacity fade due to repeated charge/discharge, and cutoff voltage considerations were folded into this pack-level specific energy metric. Pack output energy was computed from total trim shaft power and the associated electromechanical conversion efficiencies (86% for motor and speed controller).

6) **Engine:** A database of commercial-off-the-shelf internal combustion engines was collected. Engine mass and specific fuel consumption are plotted against rated engine power in Figs. 5a and 5b, respectively. The smaller engines in Fig. 5a (gray dots/line) are hobby-grade, requiring frequent maintenance. The advertised life of these engines is a few tens of hours. Aviation-grade internal combustion engines are typically operated for thousands of hours; these designs are shown as red dots and lines, and labeled as “large” reciprocating engines. In this study, the “large” engines are used for sizing power plants, with the engine mass represented with the following trend line:

$$M_{IC\text{engine}}(\text{in kg}) = 0.623P(\text{kW}) + 3.596 \quad (2)$$

The specific fuel consumption is calculated using the fitted trend line in Fig. 5b, given by

$$\text{SFC}(\text{kg/kW} \cdot \text{h}) = 5.965 \times 10^{-6}P^2 - 2.219 \times 10^{-3}P(\text{kW}) + 0.4463 \quad (3)$$

7) **Miscellaneous components:** The mass of this group was assumed to be 7.5% of takeoff mass. This accounts for additional

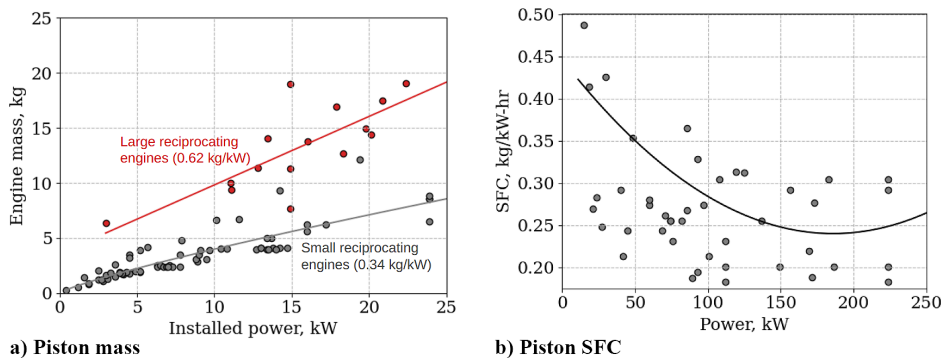


Fig. 5 Engine weight and fuel consumption trends.

systems (data recorder, transmitter, telemetry antenna, GPS/transponder, and battery health monitoring). Depending on the specific components chosen, a fixed mass estimate may be used to replace this assumption in refinement steps; for early conceptual sizing, we assume a representative fraction of takeoff mass.

2. Aerodynamic Modeling

The trim methodology and aerodynamic models for the multi-copter system are based on the authors' previous work [14], and these models were adapted to cast sizing as a formal MDO problem. The modifications to the baseline model are highlighted in this subsection.

1) **Nonlifting bodies:** A vertical download factor of 2% in hover was used for sizing, which is conservative for the tailsitter and multicopter vehicles considered. The centerbody ("fuselage"), with dimension of $0.5 \text{ m} \times 0.4 \text{ m} \times 0.4 \text{ m}$, has a maximum internal volume of 80 liters, providing sufficient capacity to accommodate aircraft systems, power plant(s) and payload. Parasitic drag in cruise was calculated using the projected frontal area of the centerbody and a drag coefficient. The centerbody drag coefficient varies with fuselage pitch attitude as

$$C_{D\text{fuselage}} = 0.1 + 0.2 \sin^3 \theta_{\text{fuselage}} \quad (4)$$

$$f_{\text{fuselage}} = C_{D\text{fuselage}} S_{\text{fuselage}} \quad (5)$$

Here, S_{fuselage} is equal to 0.27 m^2 ($0.5 \times 0.4 + 0.5\pi 0.2^2$). Spinners on prop-rotors are assumed to have a C_D of 0.3. The spinner diameter is assumed to scale proportionally with the rotor diameter.

$$f_{\text{spinner}} = \pi(0.05R)^2 C_{D\text{spinner}} \quad (6)$$

where R is the rotor radius. To account for interference effects, an additional 20% parasitic drag is added to total drag. Drag of fairings, fasteners, and protrusions was modeled as an additional 0.05 m^2 of equivalent flat-plate area. The total flat-plate area is given by

$$f_{\text{total}} = 1.2(f_{\text{fuselage}} + N_R f_{\text{spinner}}) + 0.05 \quad (7)$$

where N_R is the number of rotors.

2) **Wings:** Wing lift was predicted using sectional tables for the SD7032 airfoil, with angle-of-attack variation between -180° and 180° . Wing drag was calculated as the sum of profile drag ($C_{D0} = C_{d\text{airfoil table}} + 0.005$ to account for increased drag from rotor-wing interference) and induced drag with an Oswald efficiency factor $e = 0.8$. The effect of propeller wake on the wings was included by vectorially adding the slipstream velocity of the propeller to the freestream velocity vector on the wing. Consequently, changes in the local angle of attack over sections of the wing affected by the propwash were also included in this formulation.

3) **Rotor blades:** Rotor power was predicted using modified momentum theory. Profile power was calculated based on a mean sectional drag coefficient for the blade airfoil at $C_{d0} = 0.022$. Induced power was computed assuming uniform inflow over the rotor disk, with empirical corrections for nonideal losses ($\kappa = 1.1$). Thrust capabilities of fixed-pitch edgewise rotors in edgewise flight and accuracy verification of the rotor performance model are explained in Appendices A and B, respectively.

3. Trim

The power required in each mission segment (flight condition) was obtained by satisfying the vehicle force and moment equilibrium equations. Longitudinal trim of a winged biplane tailsitter is shown in Fig. 6, and the details of the trim problem formulation are provided in [14]. Using the SAND formulation for sizing and performance analysis obviates the need for ad hoc exterior penalty functions (to account for cases that fail to trim) and improves the generality of the framework.

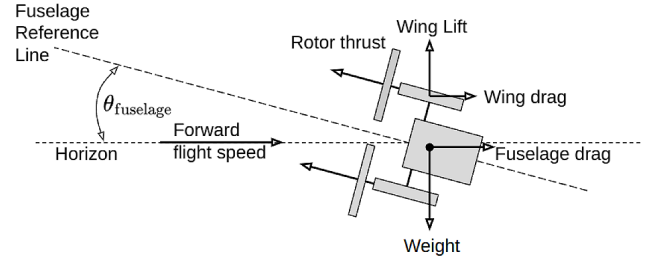


Fig. 6 Tailsitter aircraft configuration, longitudinal trim in HYDRA.

III. Results and Discussion

A. Vehicle Configurations and Mission Profile

The key advantage of distributed electric propulsion lies in the mechanical simplicity associated with multiple rotors powered by electric motors. At the weight classes of interest (100 kg takeoff mass), a low-maintenance requirement may drive toward a mechanically simple multicopter layout without rotor blade hinges or flexures. An example four-rotor layout is shown in Fig. 7a; this configuration cannot tolerate single motor failures and continue flying. Hexacopter and octacopter configurations can both continue to fly after a single motor or rotor failure; these configurations are also sized and compared to a quadcopter.

To enable energy-efficient cruise (thus lowering energy storage system weight), fixed wings are required to reduce cruise power. A biplane layout is especially advantageous because wing spars are a natural choice for rotor mounts—the wing spars carry vertical lift and associated bending moments in both hover and cruise, serving a dual purpose. An example quadrotor biplane configuration is shown in Fig. 7b. This configuration may be expanded to hexarotor biplane and octarotor biplane variants by extending the wing span to accommodate the additional rotors. The configurations considered for this work are tailsitter aircraft, with the ability to pitch the body nose-down by approximately 90° to transition between hover and cruise. Variable-pitch and variable-RPM prop-rotors enable good hover figures of merit and good cruise propeller efficiencies with the same blade shape. The cost of higher operating efficiencies for the winged configurations is paid up-front in additional parts as well as development time to fine-tune flight controls, especially during transition.

B. Comparison Between a Quadcopter and QBiT

This section compares sizing results for a quadcopter with its winged biplane tailsitter variant (QBiT) for a fixed takeoff mass (100 kg). The sizing mission profile is shown in Fig. 7c. Inbound and outbound transition energy usage was approximated using equivalent hover segments, the durations of which are also shown in Fig. 7c. Sizing is carried out assuming that the payload is not dropped at the delivery site, to account for cases where the landing pad is blocked when the UAV attempts to drop the package; this assumption results in a more conservative vehicle design because sizing is performed for the highest possible operating weight in all mission segments.

The package delivery distance is denoted by d , measured in kilometers. The following studies were carried out for this sizing mission:

- 1) Comparison of quadcopter versus QBiT for an all-electric power plant for a delivery distance $d = 25 \text{ km}$
- 2) Comparison of quadcopter versus QBiT for a piston engine/generator power plant for a delivery distance $d = 100 \text{ km}$
- 3) Comparison of quadcopter versus QBiT for a hybrid power plant for a delivery distance $d = 100 \text{ km}$
- 4) Upgrading a quadcopter to a winged biplane tailsitter (QBiT) one component at a time to identify the upgrades critical to unlocking increased payload
- 5) Understanding the impact of piston engine SFC on quadcopter and QBiT for both piston engine/generator power plant and hybrid power plant
- 6) Exploring the performance of six-rotor and eight-rotor multicopters and winged tailsitters

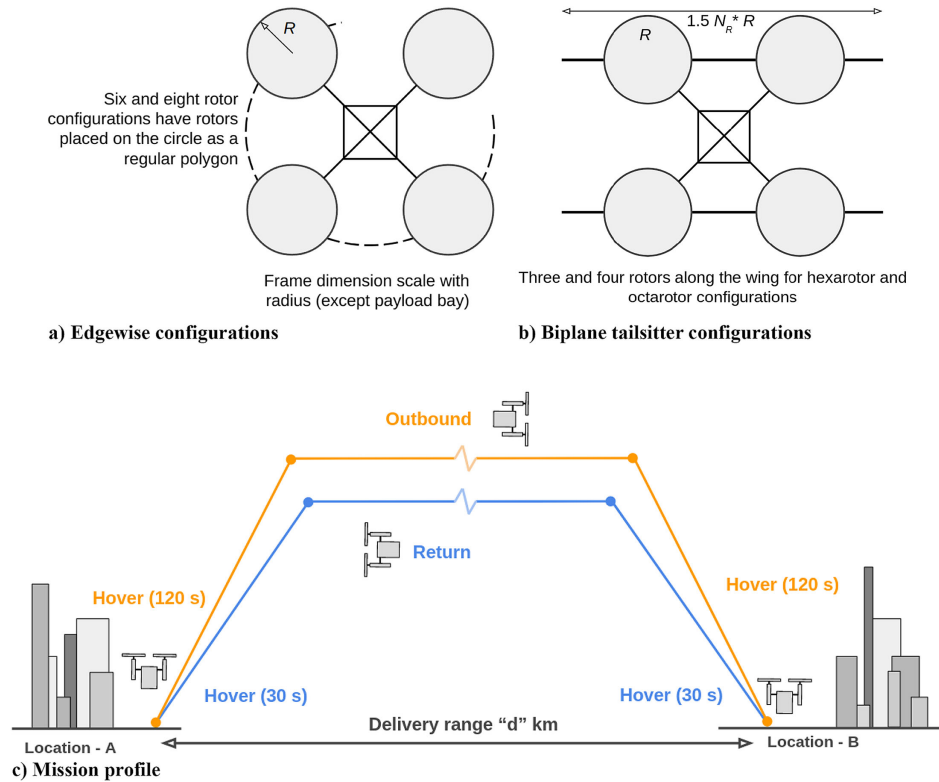


Fig. 7 Vehicle configurations of interest and chosen mission.

1. Sizing Results: Battery-Electric Powertrain

A battery-electric quadcopter and quadrotor biplane tailsitter (QBiT) were optimized to carry maximum payload weight for a delivery distance of $d = 25$ km. For both vehicles, the takeoff mass was held fixed at 100 kg. A side-by-side comparison of the key vehicle dimensions (top view in hover) is shown in Fig. 8a. The left half of the figure shows the quadcopter's rotors and support arm. The right half of the figure shows the biplane tailsitter configuration's rotors, wing, and support arm.

Table 3 shows the key weight, geometry, and performance metrics for the quadcopter and QBiT configurations. An underlined entry indicates that the optimized vehicle parameters were restricted by designer-imposed variable bounds. For instance, rotor solidity (for both optimized configurations) lies on the lower bound of 0.06, imposed to prevent minuscule rotor blade chord. The rotor RPM reduction in cruise (denoted by the ratio cruise rotor RPM/hover rotor RPM) for the tailsitter configuration also lies on the lower bound of 0.5, imposed to avoid driving the motor to regions of low (<75%) operating electromechanical efficiencies.

Figure 8b shows the power flow for the battery-electric quadcopter and QBiT. The larger rotors on the quadcopter result in slightly heavier rotor blades compared to the tailsitter. Lower hover tip speeds (and larger rotor disks) result in higher sizing motor torque and heavier motors for the quadcopter. The absence of fixed wings (and associated load paths) results in a simpler and lighter fuselage. Empty weight ratios for both optimized configurations are very similar: the difference is less than 2% of takeoff mass. However, the payload for the biplane tailsitter configuration is almost double that achieved by the optimized quadcopter. This payload increase for the biplane tailsitter is enabled by the superior cruise aerodynamic efficiency of fixed wings to produce vertical lift in forward flight. Here, cruise efficiency is quantified by the vehicle equivalent lift-to-drag ratio L/D_e (defined as weight $\times$ airspeed/rotor shaft power). The tailsitter features a vehicle L/D_e that is nearly thrice that of the quadcopter. For these delivery distances, the biplane tailsitter configuration enables higher payload-carrying capabilities.

To analyze the sensitivity of each configuration's payload-carrying capabilities to mission profile parameters, two studies were performed:

1) Design cruise speed study: Several quadcopters and biplane tailsitters were optimized for design cruise speeds varying from 30 to 100 knots, holding the delivery distance $d = 25$ km fixed. The results are shown in Fig. 9a. Each point on the two curves corresponds to a unique vehicle optimized for maximum payload and flying at the cruise speed specified on the horizontal axis. For each configuration, there is a design cruise speed at which the payload-carrying capability is maximum (55 knots for the quadcopter and 65 knots for the biplane tailsitter). Flying faster than 70 knots with the quadcopter results in a steep decrease in payload, while the tailsitter configuration shows a more benign reduction in payload (30%) at 100 knots.

2) Design delivery distance study: Several quadcopters and biplane tailsitters were optimized for design delivery distances varying from 5 to 80 km, holding the cruise speeds fixed (values shown in Table 3). The results are shown in Fig. 9b. Each point on the two curves corresponds to a unique vehicle optimized for maximum payload for the delivery distance specified on the horizontal axis. Below 10 km delivery distance, the quadcopter features a higher design payload. For delivery distances greater than 20 km, the biplane tailsitter can carry more payload. The superior cruise efficiency of the tailsitter configuration results in a shallower decrease in design payload with delivery distance compared to the quadcopter; the maximum delivery distance for a 10 kg payload is 60 km for the tailsitter and 30 km for the quadcopter.

The higher payload-carrying capability, higher cruise speeds, and longer range result in the battery-electric biplane tailsitter offering more operational utility compared to the edgewise quadcopter.

2. Sizing Results: Piston Engine Power plant with Electric Drivetrain

To overcome the range limitations of battery-powered designs, another power plant option was considered for sizing both vehicles: a hydrocarbon-burning piston engine. To distribute power to the four rotors without incurring a mechanical complexity penalty, a

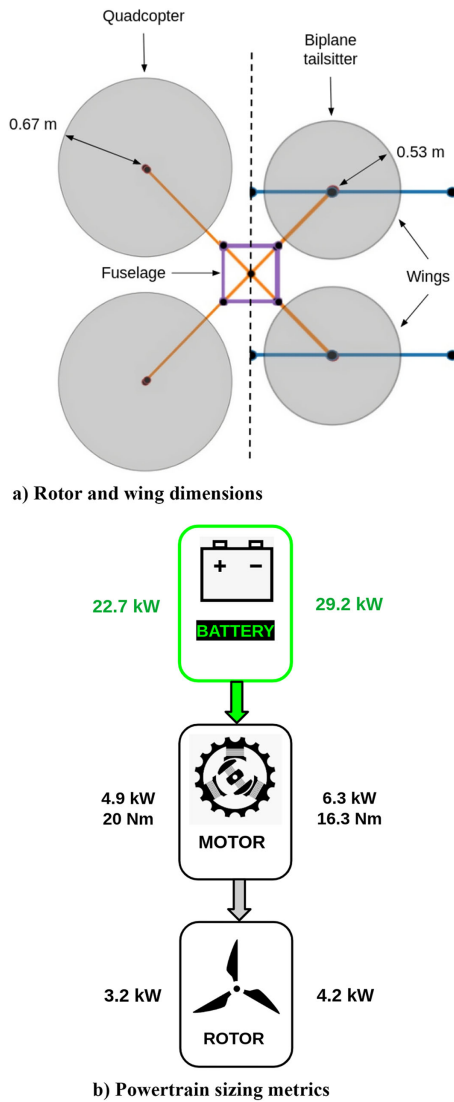


Fig. 8 Comparison of vehicle dimensions and powertrain sizing metrics for battery-electric quadcopter and QBiT.

generator/rectifier combination was included in vehicle sizing results. The engine, generator and rectifier were sized to provide $1.5\times$ their corresponding hover trim values to ensure that maneuvering flight requirements and gust rejection capabilities are retained.

The quadcopter and quadrotor biplane tailsitter were optimized to carry maximum payload weight for a delivery distance of $d = 100$ km. For both vehicles, the takeoff mass was held fixed at 100 kg, and the cruise airspeed was frozen. A side-by-side comparison of the key vehicle dimensions (top view in hover) is shown in Fig. 10a. The left half of the figure shows the quadcopter's rotors and support arm. The right half of the figure shows the biplane tailsitter configuration's rotors, wing, and support arm. Figure 10b shows sizing metrics for various powertrain components of both vehicle configurations.

Table 4 shows the key weight, geometry, and performance metrics for the quadcopter and QBiT configurations. Underlined entries indicate that the optimized vehicle parameters were restricted by designer-imposed variable bounds. As observed in the battery-electric designs, rotor solidities lie on the lower bound of 0.06, and rotor RPM reduces to 50% in cruise for the tailsitter configuration.

The optimized piston-engine-powered designs feature high empty weight fractions (67% for the quadcopter and 77% for the tailsitter). These high empty weight ratios are reflective of heavy engines and associated generators and rectifiers sized for $1.5\times$ their

Table 3 Vehicle sizing results for 100 kg battery-electric UAVs for 25 km delivery distance

Parameter	Quadcopter	QBiT
Cruise airspeed, knots	55	65
Mass		
Takeoff mass, kg	100.00	
Battery mass, kg	46.44	27.56
Empty weight, kg	30.99	32.3
Payload, kg	19.57	37.14
Common equipment, kg	3	3
Component weight		
Rotor mass, kg	5.72	4.42
Fuselage mass, kg	7.47	8.77
DC motor mass, kg	10.28	8.71
Miscellaneous, kg	7.5	7.5
Wing mass, kg	—	2.89
Rotor and wing		
Rotor radius, m	0.67	0.53
Rotor solidity	0.060	0.060
Hover tip speed, m/s	142.25	178.89
Cruise RPM ratio	0.99	0.5
Rotor thrust (hover), kg	25.49	25.55
Rotor thrust (outbound cruise), kg	25.06	2.64
Wing span, m	—	3.19
Wing aspect ratio	—	14.69
Lift characteristics		
Wing operating C_L	—	1
Vehicle L/D_e	2.93	7.89

Quantities that are underlined are bounded by side constraints.

hover trim values. The generator, rectifier, and engine constitute 33% and 38% of takeoff mass for the quadcopter and biplane tailsitter, respectively. As a consequence of these heavy power plants and electromechanical energy converters, the payload-carrying capability of the piston-engine-powered designs with electrified drivetrains is diminished compared to their battery-electric counterparts.

Another consequence of the high mass penalty of power plants is that the optimized designs feature larger rotors compared to their battery-electric counterparts. These larger rotors also require larger and heavier airframes—and wings, for the tailsitter. The combined mass fraction of motors, rotors, airframe, and wings is 26% for the quadcopter and 31% for the biplane tailsitter. These mass fractions are comparable to the 33% and 38% mass fractions for the engine, generator, and rectifier for the quadcopter and tailsitter, respectively. Therefore, the resulting rotor size is driven by the conflicting objectives of reducing power plant/powertrain weights, and reducing rotor, motor, airframe, and wing masses.

The optimized tailsitter features a cruise aerodynamic efficiency that is roughly $3\times$ that of the quadcopter; the corresponding mission fuel burn is proportionally reduced for the tailsitter. Both optimized configurations feature payload fractions of approximately 15%.

Sensitivity study: To analyze the sensitivity of each configuration's design payload to mission profile assumptions, two studies were performed:

1) Design cruise speed study: Several quadcopters and biplane tailsitters were optimized for design cruise speeds varying from 30 to 100 knots, holding the delivery distance $d = 100$ km fixed. The results are shown in Fig. 11a. Each point on the two curves corresponds to a unique vehicle optimized for maximum payload and flying at the cruise speed specified on the horizontal axis. At 50 and 55 knots, the payload-carrying capability is maximum for the quadcopter and tailsitter, respectively. The tailsitter configuration shows a very benign reduction in design payload (15–21%) as cruise speed increases from 55 to 85 knots.

2) Design delivery distance study: Several quadcopters and biplane tailsitters were optimized for design delivery distances varying from 5 to 125 km, holding the cruise speeds fixed (values shown in Table 4). The results are shown in Fig. 11b. Each point on the two curves corresponds to a unique vehicle optimized for

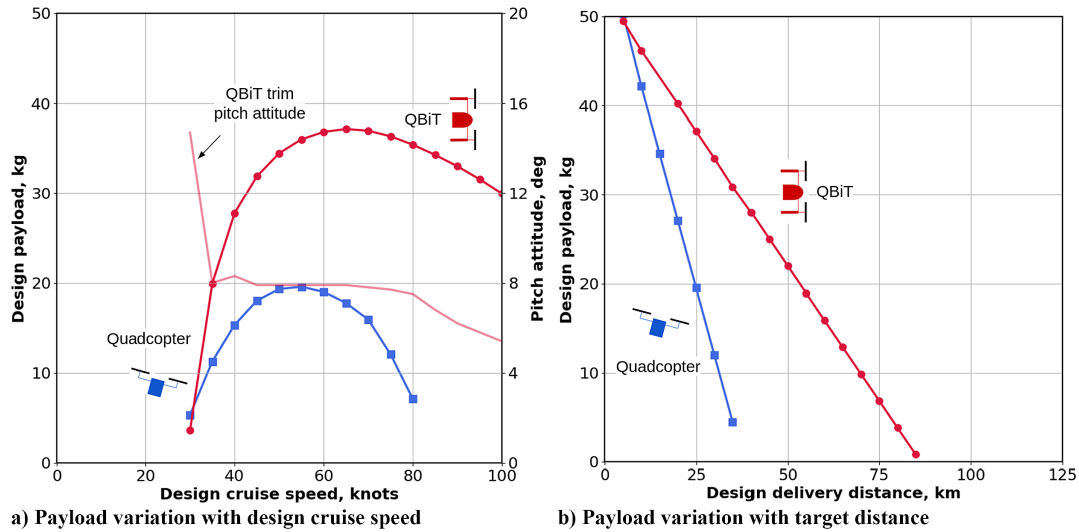
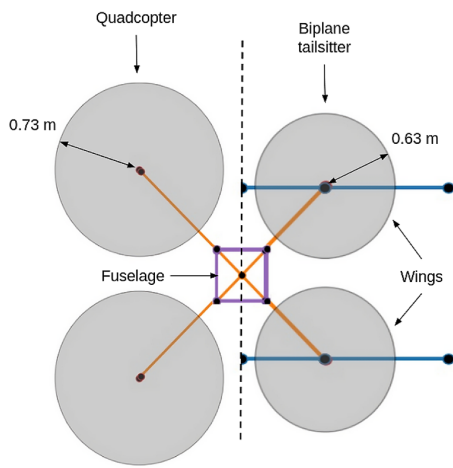
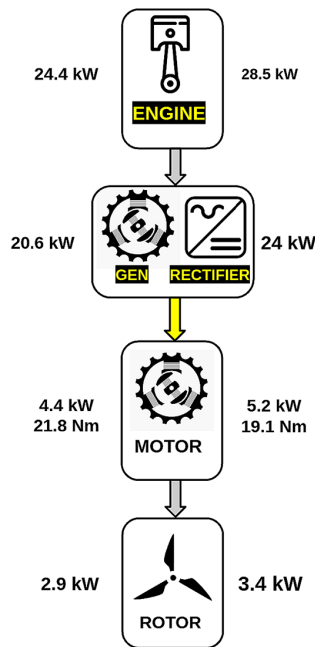


Fig. 9 Variation of payload with design cruise airspeed for quadcopter and QBiT with electric powertrain.



a) Rotor and wing dimensions



b) Powertrain sizing metrics

Fig. 10 Comparison of vehicle dimensions and powertrain sizing metrics for piston-engine-powered quadcopter and QBiT with electric drivetrains.

Table 4 Vehicle sizing results for 100 kg piston-powered UAVs for 100 km delivery distance

Parameter	Quadcopter	QBiT
Cruise airspeed, knots	50	55
Mass		
Takeoff mass, kg	100.00	
Fuel mass, kg	14.89	5.62
Empty weight, kg	66.93	76.77
Payload, kg	15.18	14.61
Common equipment, kg	3	3
Component weight		
Rotor mass, kg	6.62	5.33
Fuselage mass, kg	8.51	11.59
Engine mass, kg	18.79	21.33
DC motor mass, kg	11.1	9.92
Generator mass, kg	14.4	16.8
Miscellaneous, kg	7.5	7.5
Wing mass, kg	—	4.28
Rotor and wing		
Rotor radius, m	0.73	0.63
Rotor solidity	0.060	0.060
Hover tip speed, m/s	129.69	149.69
Cruise RPM ratio	0.99	0.5
Single rotor thrust (segment 1), kg	25.49	25.55
Single rotor thrust (segment 2), kg	24.96	2.2
Wing span, m	—	3.81
Wing aspect ratio	—	15
Lift characteristics		
Wing operating C_L	—	1.0
Vehicle L/D_e	3.07	9.19

Quantities that are underlined are bounded by side constraints.

maximum payload for the delivery distance specified on the horizontal axis. With the piston engine power plant, the quadcopter features a higher design payload than the tailsitter for less than 100 km delivery distance. For delivery distances greater than 105 km, the biplane tailsitter can carry more payload. The piston engine (with its associated electric drivetrain) pays an empty weight penalty “up-front” and enables longer delivery distances for both the quadcopter and tailsitter through use of energy-dense fuel. However, these designs (featuring fuel-burning engines) reduced payload for short (≤ 25 km) delivery distances because of their higher empty weights.

Below 25 km delivery distance (60 km for tailsitter), a piston-engine-powered design carries less payload than its battery-electric counterpart. For longer delivery distances, fuel-burning engines feature higher payload fractions and offer more utility.

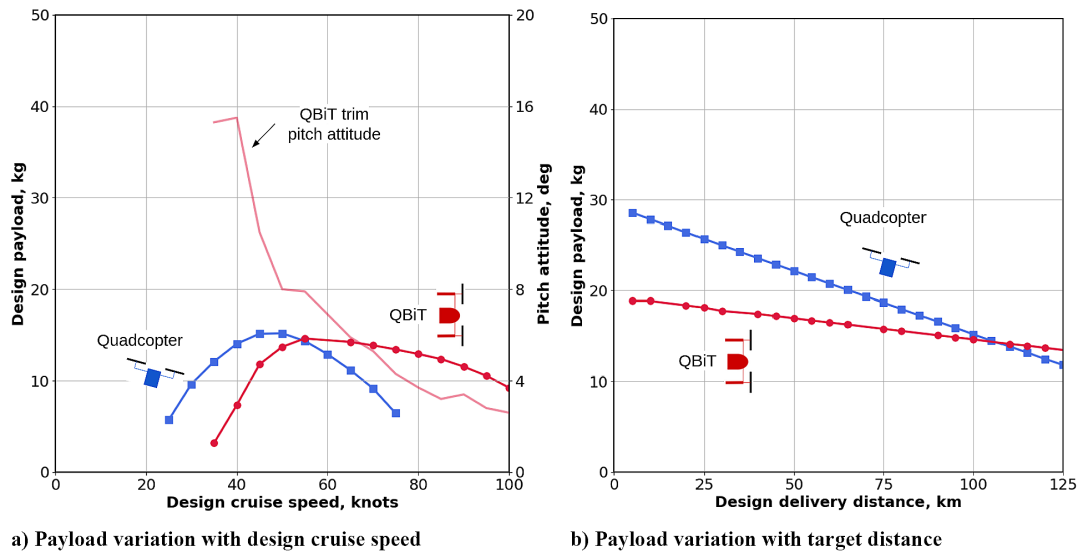


Fig. 11 Variation of payload with design cruise airspeed for quadcopter and QBiT with piston engine powertrain.

C. Case for a Series Hybrid Power plant

The battery pack performs the equivalent combined functions of a conventional power plant (engine), its associated energy converters (generator and rectifier) and its energy storage system (fuel tank and fuel). Put simply, the battery pack is both a power source and an energy source. In mass-normalized terms, the battery may be compared to a power plant using a “kW/kg” metric and to hydrocarbon fuel using a “Wh/kg” metric to identify lightweight options for both power production and energy storage.

Applying a 12E limit (5 minutes operation), the specific power of a battery producing DC power is approximately 2 kW/kg, or four times the specific power of a piston engine/generator/rectifier combination. Producing DC power for 5 minutes with a battery pack requires a system that weighs only a quarter of the mass required to produce the same power using a piston engine, generator, and rectifier.

For the same mass budget, a battery pack rated at 158 Wh/kg can provide approximately 5% of the electrical energy output of a fuel-burning piston engine. We may conclude, based on this first-order analysis, that the battery is a preferable *power source* for short durations but a significantly heavier energy storage solution than hydrocarbon fuel. In other words, *the battery is a lighter power source compared to a fuel-burning engine but a heavier energy storage solution than hydrocarbon fuels*. These attributes render a battery pack ideal for producing *short-duration* power, while fuel-burning engines are ideal for providing steady (relatively lower) power for long durations. Using this property of a battery, we may reduce the power system mass on a VTOL aircraft through hybridization.

Typically, the engine in a low-speed VTOL aircraft is sized for hover power (plus margins), because hover is the most power-demanding flight condition in the mission profile. Cruise power is less than hover power for such platforms. However, if the engine were sized to provide cruise power only (and the remaining power required for hover and transient maneuvers was supplied by a fast-discharge battery), the overall power system mass may be reduced. For a 100 kg UAV with a rotor hover disk loading of 20 kg/m², the mass of the piston engine, generator, and rectifier as well as the mass of the boost battery are plotted for various values of cruise power in Fig. 12. A hover shaft power of 4.25 kW per rotor was assumed in these calculations.

Figure 12 shows that the power system mass savings are significant when cruise power is much lower than hover power (20–30%). These cruise power to hover power fractions are typical of VTOL aircraft with fixed wings to provide lift compounding in combination with high-efficiency propellers. Such aerodynamic performance targets are hard to achieve with configurations featuring edgewise rotors. For example, the optimized quadcopters feature L/D_e values of approximately 3 (i.e., cruise power is approximately 70% of hover power),

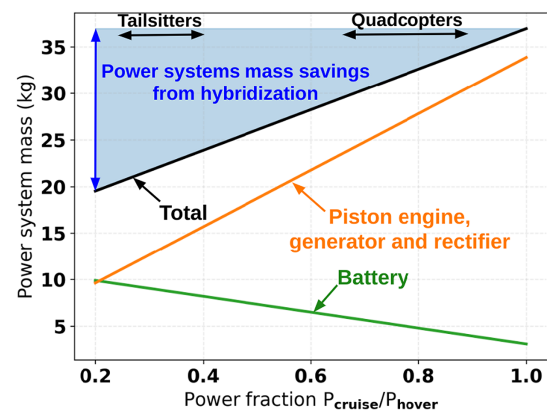
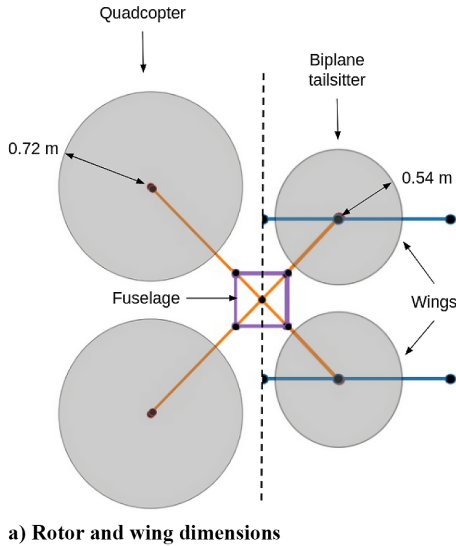


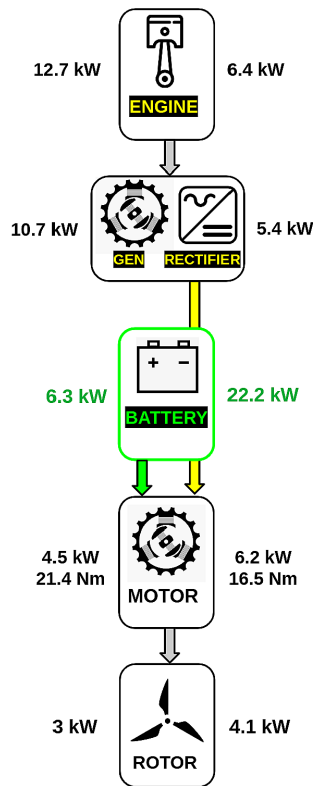
Fig. 12 Power plant mass savings obtained from a series hybrid compared to piston engine/generator/rectifier power plant.

while the optimized biplane tailsitters feature L/D_e values of 8–9 (cruise power is approximately 25% of hover power). For vehicles with higher cruise aerodynamic efficiencies (and consequently, larger differences between hover and cruise power), a hybrid-electric architecture offers weight reductions of (approximately) 15% of takeoff mass. For a quadcopter with lower cruise aerodynamic efficiencies, hybridization may still enable modest weight reduction (6% of takeoff mass). On a hybrid VTOL (e.g., tailsitter) configuration, these weight reductions justify the use of a more complex dual-power plant series-hybrid powertrain. The quadcopter and quad-biplane tailsitter were subsequently reoptimized with the dual-power plant architecture to quantify the weight reduction benefits offered by hybridization; key vehicle metrics are shown in Table 5. A side-by-side comparison of the key vehicle dimensions (top view in hover) is shown in Fig. 13a, the left half shows the quadcopter’s rotors and support arm and the right half shows the biplane tailsitter configuration’s rotors, wing, and support arm. Figure 13a shows sizing metrics for various powertrain components of both vehicle configurations. For vehicles featuring the series hybrid powertrain, the engine, generator and rectifier are sized for 110% cruise power, while the battery is sized with an additional 20% energy budget to ensure that intermittent high power may be provided on demand for maneuvering flight.

The optimized vehicles (Table 5) feature rotor and wing characteristics that are very similar to their single-power plant (piston engine) powered counterparts—larger quadcopter versus smaller tailsitter discs, slender blades (6% solidity), reduced quadcopter tip speeds in hover, and cruise aerodynamic efficiencies roughly similar to the designs shown in Table 4 (L/D_e of 3 for the quadcopter at 50 knots and 8 for the tailsitter at 65 knots).



a) Rotor and wing dimensions



b) Powertrain sizing metrics

Fig. 13 Vehicle dimensions and powertrain sizing metrics for series-hybrid quadcopter and QBiT.

The key benefit of power plant hybridization is evident in the empty weight reductions for both vehicles (65–75% for piston engine vs 45–50% for series-hybrid). These empty weight reductions are primarily driven by weight reductions for the engine, generator and rectifier. For the single-power plant case (Table 4), all powertrain components—piston engine, generator, rectifier, and motor/ESC—were sized to accommodate 150% of hover power to enable gust rejection and maneuvering flight. For the series-hybrid dual-power plant case (Table 5), only the ESC and motors were sized for 150% hover power; the battery pack provides the excess power for transient flight conditions. This shift in the target sizing power condition for the engine—110% cruise instead of 150% hover—is the single biggest driver for empty weight reduction. Introducing a series-hybrid powertrain roughly doubles the payload for a quadcopter from 15% to 30%

Table 5 Vehicle sizing results for 100 kg MGTOW and 100 km delivery distance with series-hybrid dual-power plant architecture

Parameter	Quadcopter	QBiT
Cruise airspeed, knots	50	65
Mass		
Takeoff mass, kg	100.00	
Battery mass, kg	2.14	8.9
Fuel mass, kg	11.73	4.69
Empty weight, kg	52.15	44.03
Payload, kg	30.98	39.48
Common equipment, kg	3	3
Component weight		
Rotor mass, kg	6.41	4.45
Fuselage mass, kg	8.26	8.95
Engine mass, kg	11.53	7.57
DC motor mass, kg	10.92	8.78
Generator mass, kg	7.52	3.77
Miscellaneous, kg	7.5	7.5
Wing mass, kg	—	2.9
Rotor and wing		
Rotor radius, m	0.72	0.54
Rotor solidity	0.060	0.060
Hover tip speed, m/s	132.27	176.63
Cruise RPM ratio	0.99	0.5
Single rotor thrust (hover), kg	25.49	25.55
Single rotor thrust (cruise), kg	24.99	2.62
Wing span, m	—	3.23
Wing aspect ratio	—	15
Power plant		
Power fraction (hover)	0.3	0.74
Lift characteristics		
Wing operating C_L	—	1
Vehicle L/D_e	3	7.95

Quantities that are underlined are bounded by side constraints.

(compared to a piston engine power plant); for a tailsitter, the payload increase is even more dramatic (15–40%).

Sensitivity study: The sensitivity of each configuration's design payload to sizing mission profile assumptions was studied through two parametric studies:

1) Design cruise speed: Several quadcopters and tailsitters were optimized for design cruise speeds varying from 30 to 100 knots, holding the delivery distance $d = 100$ km fixed. The results are shown in Fig. 14a. Each point on the two curves corresponds to a unique vehicle optimized for maximum payload and flying at the cruise speed specified on the horizontal axis. The payload-carrying capability is maximum for the quadcopter at 55 knots and for the tailsitter at 60–65 knots, respectively. The fraction of hover power supplied by the battery is also shown for both configurations using semi-transparent lines (without markers) on a second vertical axis; these lines are annotated as “Quad power fraction” and “QBiT power fraction” in the plot. At the maximum payload design cruise speeds, the quadcopter draws 32% of its required hover power from a battery; the tailsitter draws more than 80% of its hover power from a battery. This ratio is indicative of the differences between hover and cruise power required for both platforms and a key driver for powertrain weight reductions.

2) Design delivery distance study: Several quadcopters and biplane tailsitters were optimized for design delivery distances varying from $d = 5$ km to $d = 125$ km at 50 knots for the quadcopter and 65 knots for the tailsitter. The results are shown in Fig. 14b. Each point on the two lines corresponds to a unique vehicle that was optimized for maximum payload for the delivery distance specified on the horizontal axis. The series-hybrid tailsitter features an incrementally higher design payload than its quadcopter counterpart for delivery distances up to $d = 25$ km. As the delivery distance increases to 125 km, the tailsitter's higher cruise efficiency affords almost 10 kg more payload than the quadcopter.

Hybridizing the quadcopter enables a lightweight powertrain and increases its payload-carrying capability for short delivery distances (29 kg increasing to 42 kg at $d = 5$ km) as well as for longer delivery

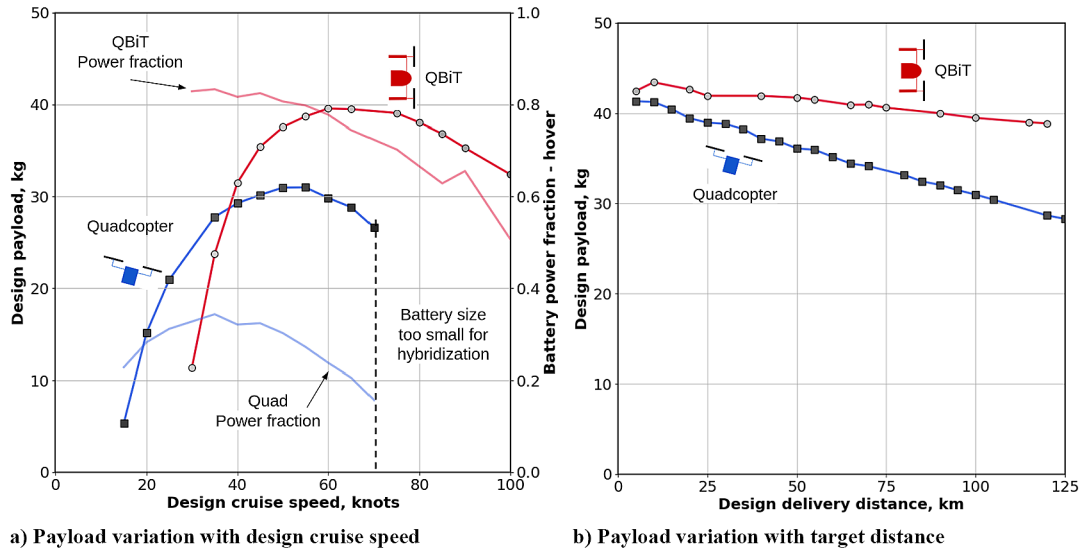


Fig. 14 Variation of payload with design cruise airspeed for quadcopter and QBiT with hybrid powertrain.

distances (11 kg increasing to 29 kg at $d = 125$ km). These gains are higher still for the tailsitter (20–42 kg at $d = 5$ km, and 13–38 kg at $d = 125$ km). Hybridizing a tailsitter features consistently higher design payloads (than a quadcopter) for both short and long delivery distances; the “cost” for this benefit lies in the complexity of developing a robust powertrain to operate in several vehicle orientations, especially during transition between hover and cruise pitch attitudes.

D. Incremental Development of Hybridized Tailsitter

The previous sections highlighted the advantages of hybridizing a biplane tailsitter and quantified the enormous increase in payload-carrying capability unlocked by the dual-power plant architecture. In this section, we propose an incremental development approach for maturing different aspects of the hybridized tailsitter through two sequential upgrades, starting from a suboptimal (but well-understood) quadcopter configuration. The upgrades are as follows:

- 1) **Configuration update:** converting a quadcopter to a biplane tailsitter
- 2) **Power plant hybridization:** change battery-electric power plant to series-hybrid on a tailsitter

The first upgrade (configuration update) involves the addition of fixed wings and collective pitch change mechanisms for the rotor blades, as well as replacing rotor blades, hubs, and motors. The battery-electric quadcopter may be converted to a tailsitter without any fundamental changes to the airframe. The second upgrade (hybridization) involves replacement of a large battery pack with

an engine, generator, rectifier, and smaller “boost” battery, retaining the “common” airframe and rotors, hubs, motors, and fixed wings from the first upgrade. The process for determining key attributes for the three vehicles is shown in Fig. 15. The specific steps in the incremental transformation approach are detailed below:

- 1) **Size-optimized hybrid-electric tailsitter:** This configuration is shown in Table 5. All subsequent configurations retain the same rotor radius (0.54 m).
- 2) **Sizing and optimizing battery-electric quadcopter:** These are performed with the same radius as the optimized tailsitter (0.54 m). The delivery distance is adjusted to retain the same payload capacity as the optimized hybrid tailsitter (39.4 kg).
- 3) **Choosing “common” airframe:** The airframe tube sizes of the optimized quadcopter (from step 2) are compared to their counterparts from the optimized tailsitter (from step 1). The larger dimension for each unique tube type is chosen for the “common” airframe. This airframe can be used with either the quadcopter (step 2) or the tailsitter (step 1), because it can withstand operating loads from both vehicles.
- 4) **Battery-electric tailsitter:** It retains all key features of the optimized hybrid tailsitter (from step 1). Common key features include rotor radius, rotor solidity, rotor tip speeds, wing dimensions, and operating payload.

The key attributes of the three vehicles are shown in Table 6.

- 1) **Battery-electric quadcopter:** It is the starting point for this incremental upgrade. The “common” airframe is used to construct this configuration. The delivery distance for the suboptimal electric-powered quadcopter is 11.5 km. The optimized battery-electric

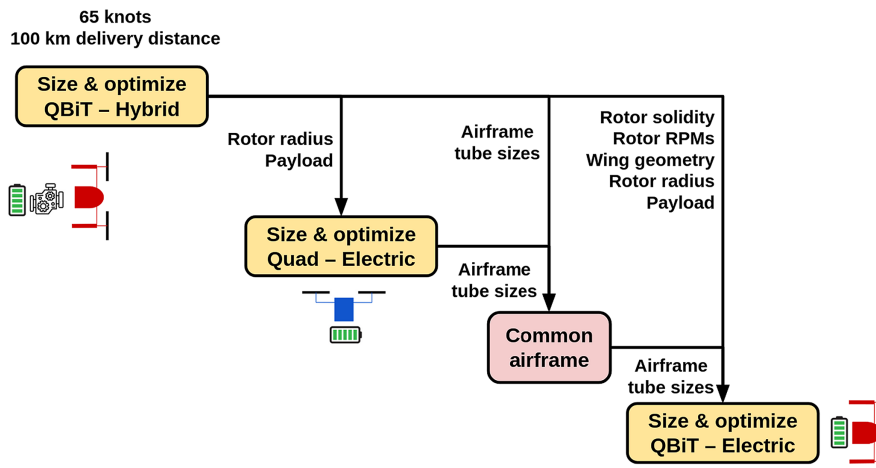


Fig. 15 Workflow to incrementally transform a quadcopter into a hybrid-electric tailsitter.

Table 6 Vehicle sizing results for 100 kg MGTOW carrying 39.4 kg payload: retrofitting quadcopter in QBiT architecture

Parameter	Quadcopter: electric	QBiT: electric	QBiT: hybrid
Cruise airspeed, knots		65	
Delivery distance, km	11.5	21	100
Mass			
Takeoff mass, kg		100.00	
Common equipment, kg		3	
Payload, kg		39.4	
Empty weight, kg	26.9	32.6	43.9
Battery mass, kg	30.6	24.9	8.9
Fuel mass, kg	—	—	4.7
Component weight			
Rotor mass, kg	4.7		4.5
Fuselage mass, kg	5.7		8.9
DC motor mass, kg	9		8.8
Wing mass, kg	—		2.9
Engine mass, kg	—		7.6
Generator mass, kg	—	—	3.8
Rotor and wing			
Rotor radius, m		0.54	
Rotor solidity	0.067		0.060
Hover tip speed, m/s	168		177
Cruise RPM ratio	1		0.5
Single rotor thrust (hover), kg	25.5		25.6
Single rotor thrust (cruise), kg	25.1		2.62
Wing span, m	—		3.23
Wing aspect ratio	—		15
Power plant			
Power fraction (hover)	—	—	0.74
Generator sizing power, kW	—	—	5.4
Engine sizing power, kW	—	—	6.4
Power			
Hover shaft power, kW	4.03		4.1
Motor (ESC) sizing power, kW	6		6.2
Motor sizing torque, Nm	17		16.5
Lift characteristics			
Wing operating C_L	—		1
Vehicle L/D_e	2.6		7.9

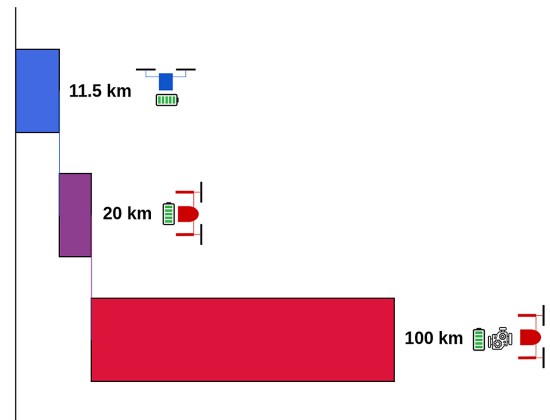
Quantities that are underlined are bounded by side constraints.

quadcopter for an 11.5 km delivery distance (see Fig. 9b) features incrementally higher but comparable payload-carrying capacities (42 kg vs 39.4 kg).

2) **Battery-electric biplane tailsitter:** It is the next configuration in the incremental upgrade sequence. The suboptimal quadcopter (with the “common airframe”) is fitted with a wing, new rotor blades, hubs, and motors. The corresponding delivery distance is 21 km for the same payload (39.4 kg). The optimized battery-electric tailsitter features near-equal payload (40 kg vs 39.4 kg) for the same delivery distance; see Fig. 9b.

3) **Series-hybrid biplane tailsitter:** It is the final step in the incremental upgrade process. The large battery in the battery-electric tailsitter is replaced with a hybrid powertrain while maintaining the “common” airframe. The delivery distance is 100 km for the same payload (39.4 kg).

The delivery distances for all three vehicles are visually depicted in Fig. 16. Adding fixed wings to a suboptimal compact quadcopter almost doubles the operating range; hybridizing the power plant enables a further fourfold increase in delivery distance. Further, the comparisons highlight a key finding: using the common airframe and designing vehicles for the incremental upgrade sequence does not noticeably diminish payload-carrying capabilities of any of the aircraft configurations as compared to their optimized counterparts. Using the incremental development approach, the shorter-range battery-electric multicopter and winged tailsitter configurations may also be matured on the path to developing a longer-range hybrid-electric tailsitter aircraft. Learnings from the battery-electric aircraft development efforts (e.g., maturing flight control laws and regulating rotor/wing lift share in transition) may be directly ported and applied to the hybrid-electric tailsitter configuration.

**Fig. 16** Delivery distances for the aircraft configurations in the incremental upgrade sequence from battery-electric quadcopter to hybrid-electric tailsitter.

E. Impact of SFC on Mission Performance

Figure 17 shows power plant power output versus mission duration. Vehicles with a hybridized power plant (i.e., piston/generator + boost battery) feature a near-constant engine power output over the sizing mission profile. For these hybrid vehicles, the impact of SFC variation with throttle setting is negligible, because changes in demanded engine power are marginal, driven only by operating weight changes from fuel burn (a weak effect for the cases analyzed).

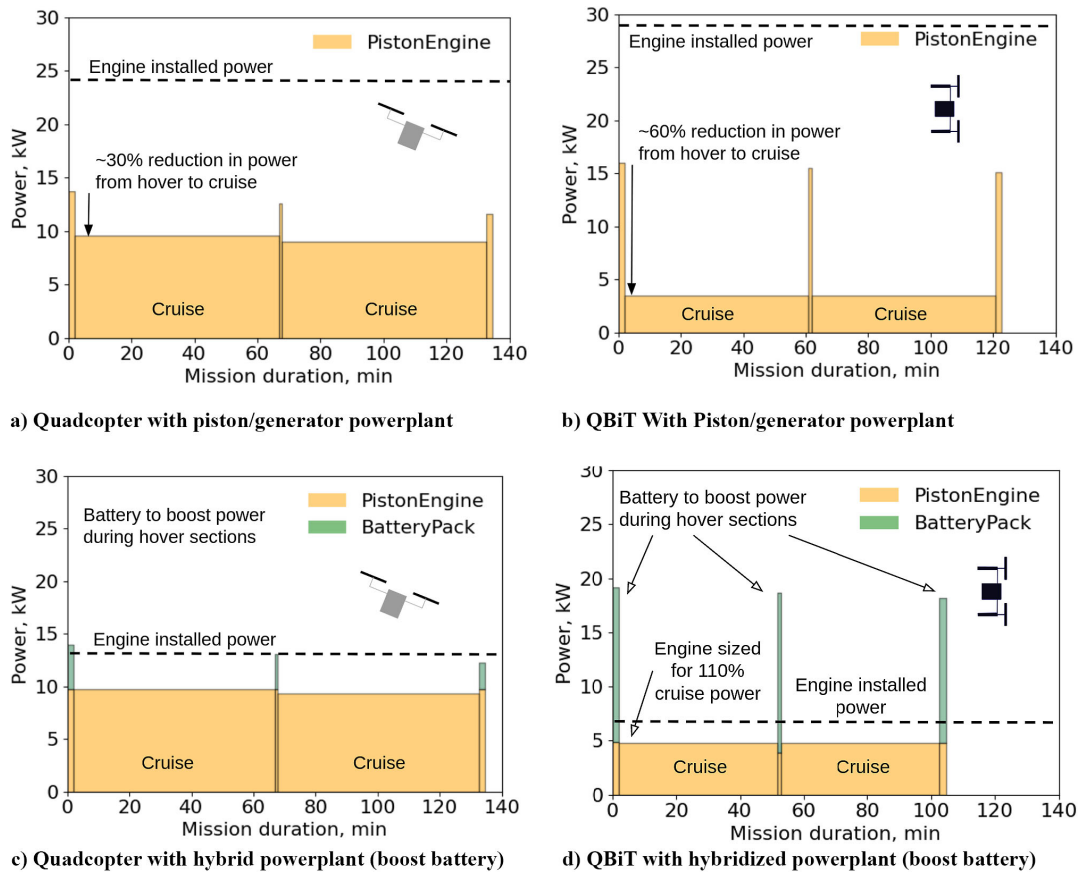


Fig. 17 Power plant power output vs mission duration for quadcopter and QBiT.

For the vehicles featuring a piston/generator power plant without a boost battery, engine power output in cruise is lower than during hover. In cruise, the quadcopter requires roughly 70% of hover power, while the QBiT requires 22% of hover power. For the tailsitter vehicles, SFC variation with throttle setting is indeed a strong sizing consideration, as noted by Arace and Datta [7].

For the vehicles without a boost battery, perturbation studies were conducted to identify the impact of engine SFC on payload. In these studies, engine SFC was increased by a factor (10% and 20%) across the *entire* mission profile (as an approximation to changing SFC only in cruise), and vehicles were resized. The quadcopter exhibits a 0.95% decrease in payload for every 1% increase in SFC. The corresponding metric for the QBiT is 0.37% decrease in payload for every 1% increase in SFC.

F. Multirotor Comparison

In this section, six-rotor and eight-rotor multicopters and biplane tailsitter vehicles are sized with a hybrid powertrain to identify whether they offer any performance or utility advantages over their four-rotor counterparts. The methodology for sizing the six-rotor and eight-rotor designs largely follows that used for the four-rotor designs, with the following key modifications:

- 1) For the six-rotor aircraft, motors are sized to provide $1.7\times$ trim hover torque.
- 2) For the eight-rotor aircraft, motors are sized to provide $1.5\times$ trim hover torque.

These ratios of sizing torque to hover torque were set so that the vehicle still retains torque margin to satisfy transient torque demands during maneuvers following single motor failure. The sizing multiplier relative to hover thrust/torque was chosen so that when one rotor fails (and the diagonally opposite rotor is throttled down for hover trim), the remaining $N-2$ rotors can provide sufficient rotor thrust (plus a 20% maneuver margin) to continue flying in a controlled manner. The sizing multiplier is set as

$N/(N-2) + 0.2$, where N is the number of rotors. For the eight rotors, this factor is approximately 1.5 (1.53, to be precise), while for the six rotors, the factor is 1.7. The optimized vehicles are shown in Table 7. The multicopter configuration and tailsitter configuration exhibit maximum payload around 50 and 65 knots, respectively. The cruise airspeed considered in this section is the same as in Table 5.

As the number of rotors increases, the radii decrease for both the multicopters and the tailsitters. Tip speeds for the optimized hexacopter and octacopter are similar to those of the optimized quadcopter. Wing properties are near-identical for all tailsitter aircraft (maximum aspect ratio and near-identical span). Vehicle equivalent lift-to-drag ratios (L/D_e) for the tailsitters are also comparable to the optimized QBiT configuration (8–8.2).

The multicopters feature higher empty weight fractions (50–55%) compared to the tailsitters (42–44%) because of heavier airframes, engines, and generators. In hover, 30% of the power demanded by the motors is supplied by batteries in the multicopters, driving up engine sizing power and fuel burn. For the tailsitters, nearly 80% of the hover power demand is supplied by the battery, enabling a lighter engine, generator, and rectifier and 60% lower fuel burn compared to the multicopters. The tailsitters feature a payload mass fraction of approximately 40%; this metric is approximately 26% for the multicopters.

Compared to a quadcopter, the hexacopter and octacopter (edge-wise) platforms feature near-identical payload-carrying capabilities. While each rotor's and motor's sizing, thrust, torque, and power are reduced as the number of rotors increases, the higher part count (for more rotors) results in comparable empty weight fractions. The same engine, generator, and rectifier could feasibly be ported from the common-airframe QBiT (shown in Table 6) to the optimized HBiT and OBiT platforms in Table 7, allowing for fewer unique parts. The key differences between the hybridized QBiT, HBiT, and OBiT with dual power plants are the airframe (more rotor mounts)

Table 7 Vehicle sizing results for 100 kg MGTOW on a ferry run returning to origin: 200 km total distance

Parameter	Hexarotor	Octarotor	HBiT	OBiT
Cruise airspeed, knots	50		65	
Mass				
Takeoff mass, kg	100.00			
Battery mass, kg	2.2	2.3	10.16	10.84
Fuel mass, kg	12.1	13.1	4.6	4.6
Empty weight, kg	55.5	56.6	43.2	42.44
Payload, kg	27.2	24.9	38.98	39.14
Common equipment, kg	3	3	3	3
Component weight				
Rotor mass, kg	6.1	5.6	5	5.6
Fuselage mass, kg	10.65	13.95	6.74	7.04
Engine mass, kg	11.85	12.6	7.5	7.4
DC motor mass, kg	11.5	8.4	9.6	7.6
Generator mass, kg	7.82	8.5	3.7	3.6
Miscellaneous, kg	7.5	7.5	7.5	7.5
Wing mass, kg	— —		3.2	3.7
Rotor and wing				
Rotor radius, m	0.57	0.46	0.37	0.29
Rotor solidity	0.060	0.060	0.089	0.12
Hover tip speed, m/s	137.1	146.2	170.5	161.7
Cruise RPM ratio	0.99	1	0.5	0.5
Single rotor thrust (segment 1), kg	16.99	12.74	17.04	12.79
Single rotor thrust (segment 2), kg	16.66	12.49	1.72	1.26
Wing span, m	— —		3.35	3.58
Wing aspect ratio	— —		15	15
Power plant				
Power fraction (Hover)	0.307	0.3	0.78	0.795
Lift characteristics				
Wing operating C_L	— —		0.94	0.83
Vehicle L/D_e	2.9	2.66	8.04	8.18

Quantities that are underlined are bounded by side constraints.

and motors/ESCs; as the number of rotors increases, a larger number of lower-torque motors and lower-power ESCs are required. The availability—or lack—of powertrain components (e.g., compact high-torque motors) may drive the choice of aircraft configuration, given that hybridized four-, six-, and eight-rotor designs feature near-identical capabilities.

IV. Conclusions

This paper presented a methodology for sizing hybrid-electric VTOL UAV configurations and explored the advantages of using different combinations of vehicle configurations and powertrain architectures. The following conclusions may be drawn:

1) A winged VTOL UAV offers significant advantages in reducing airframe drag and cruise rotor power over a multicopter configuration. The improvement in cruise aerodynamic efficiency directly translates to greater delivery distances for the winged tailsitters.

2) A series-hybrid powertrain for a winged VTOL UAV affords power system weight reductions (lighter engine, generator, and rectifier) and larger payload-carrying capabilities compared to a series-hybrid multicopter. This difference is driven by the operating cruise aerodynamic efficiencies of the winged tailsitter being nearly 3× those of a multicopter.

3) The presence of the boost battery results in a more complex powertrain architecture but enables higher payload, reduced fuel burn, and smaller engines/generators/rectifiers. The sizing results for the series-hybrid have not taken credit for these volumetric requirements, and so the predicted payload improvements for the tailsitters are a conservative estimate.

4) Shorter-range battery-electric multicopter and winged tailsitter aircraft may be matured on the path to developing a longer-range hybrid-electric tailsitter aircraft; an incremental upgrade approach was proposed in the text. These battery-electric quadcopter and tailsitter aircraft (which are compatible with the incremental upgrade approach) feature payload-carrying capabilities that are comparable to their fully optimized counterparts. The

incremental upgrade approach therefore offers the potential to mature all three aircraft configurations with maximal use of common parts.

5) The impact of engine SFC variation with throttle setting is negligible for the case of hybrid vehicles (multicopters or winged tailsitters) as changes in demanded engine power across mission segments are marginal, whereas a piston/generator power plant without a boost battery may exhibit SFC variation with throttle setting, with the tailsitters being impacted to a higher degree compared to multicopters. However, a sensitivity study was conducted to ensure that the assumption of a constant SFC is still valid for the cases studied.

6) Within the multicopter configurations, the hexacopter presents a practical “low development threshold” solution in its ability to tolerate a single rotor/motor fault and continue flying with the minimum number of rotors. The hexacopter and octacopter platforms feature near-identical payload-carrying capabilities when compared to a quadcopter. The multicopters (quadcopter, hexacopter, and octacopter) feature higher empty weight fractions compared to the tailsitter (QBiT, HBiT, and OBiT), and tailsitters also feature 60% lower fuel burn due to reduced cruise power demand. Tailsitter aircraft feature a higher payload mass fraction compared to multicopters for the chosen mission profile.

Appendix A: Thrust Capabilities of Fixed-Pitch Edgewise Rotors in Edgewise Flight

During sizing, we assumed that rigid rotors in multicopter configurations can produce the demanded trim thrust at vehicle forward airspeeds of up to 100 knots and at several operating tip speeds. To verify this assumption, a representative two-bladed rotor was analyzed at several combinations of rotor shaft tilt, forward speed, rotor RPM, and root collective pitch. The analysis was performed with a comprehensive analysis tool [20] using a blade element model of the rotor blades, sectional airfoil tables, and a three-state linear inflow model [21]. A rotor of radius 0.67 m was assumed with a 10° nose

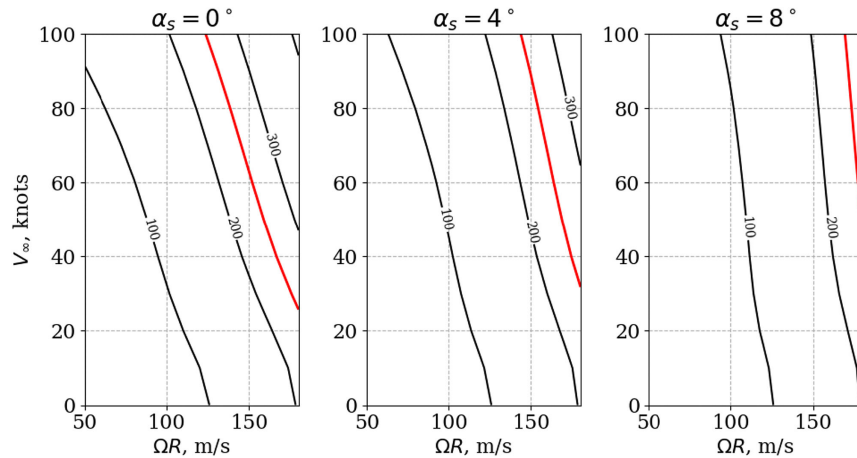


Fig. A1 Isocontours of rotor thrust for various forward flight speed, shaft angle, and hover tip speed.

down linear twist and a 2:1 linear taper ratio with a geometric rotor solidity of 0.06.

Figure A1 shows the contour lines of rotor thrust (N) for various forward flight speeds and rotational tip speeds (ΩR); these contours were generated for three different shaft tilt angles α_s : 0 deg, 4 deg forward tilt, and 8 deg forward tilt. The red line is the isocontour of 250 N, the typical value for hover rotor thrust (and notional cruise rotor thrust) in a 100 kg quadcopter. While maximum thrust generated by a rotor reduces as the shaft is tilted forward, the 250 N target thrust value, shown as a red line, is well within the lifting limits of the rotor. The contours show that the thrust targets for an edgewise rotor in a multicopter may be generated without encountering static stall.

Appendix B: Accuracy Verification of the Rotor Performance Model

To evaluate the accuracy of the simplified rotor aerodynamic performance model used in sizing, free wake predictions of rotor power (trimmed to target thrusts corresponding to hover and cruise operating points) were generated for both the edgewise quadcopter's rotor and the QBiT's prop-rotor and compared to the simplified model used in sizing. For an edgewise rotor with linear twist and no taper operating in forward flight, the detailed performance model predicts 3% higher power compared to the simplified model. Similarly, in propeller mode, free wake predictions of rotor power are 5% higher than the simplified model for the QBiT's prop-rotor for a blade with linear twist and no taper. Hover figures of merit for both rotors are within 1% of the simplified model. The gap in rotor power predictions could be closed with additional fine-tuning of twist and chord distributions along the rotor blades.

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